

FPGA-based Broadband Current Control of a Linear Motor with Class-G Power Amplifiers

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Abstract—In this paper, a broadband current control system using a FPGA and Class-G power amplifiers is realized. The current controllers with general PWM inverter, which is classified as Class-D power amplifier, does not operate in case that bandwidth of current controllers is enlarged due to PWM switching operations. On the contrary, the current control system using Class-G power amplifiers is able to be operated normally, even where the bandwidth is set as 25100 rad/s ($\simeq 4\text{kHz}$). By the numerical simulations and the experiment, validity of the current controllers with Class-G power amplifiers is verified. The broadband current control realized in this paper is useful for servo systems in the field of fast motion control and real-world haptics.

I. INTRODUCTION

In the field of industrial electronics and applications, servo systems such as position control, velocity control, impedance control, and force control are used extensively. Moreover, wide variety of control techniques has been much studied in the research area. For example, a high-performance position servo system of a induction motor was realized by sliding mode control [1], [2]. A position controller for driving a switched reluctance motor has been proposed in [3]. Furthermore, a technique of force control [4], [5] was studied as well as position control. Besides, bilateral control, which is capable of transmitting haptic and tactile sensations by using servo systems, has been also researched [6], [7]. A motion-copying system for storing and reproducing motion of a human operator was proposed [8], [9].

The issue of the bilateral control and the motion-copying system is bandwidth of control systems. In [10], bandwidth of haptic organs of a human body is up to 400 Hz. However, control bandwidth of servo systems for industrial applications is narrow, compared with haptic organs. In concrete terms, bandwidth of force control and position control is restricted by the current controllers, which is implemented as inner loop. Because, PWM inverters for driving electric actuators, output switching noise. Also, PWM carriers affect a control period of servo systems. Due to a power loss and switching time of devices such as MOSFET and of IGBTs, a frequency of PWM carriers cannot be set as higher. Hence, the bandwidth of the current control is limited.

Therefore, this paper utilizes Class-G power amplifiers [11]–[16] for high-speed servo systems. The Class-G power amplifier is sort of linear amplifier, however, power efficiency is

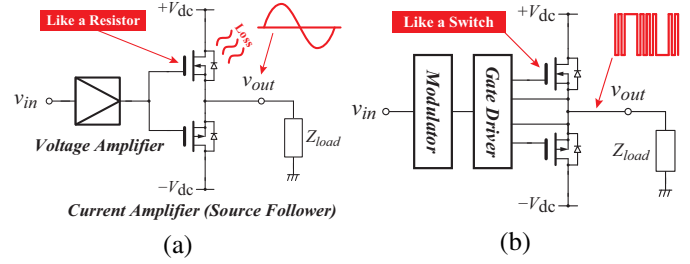


Fig. 1. (a) Class-B power amplifier. (b) Class-D power amplifier.

high as with the PWM inverter, which is classified as Class-D power amplifier. Also, current noise is able to be reduced. It is also important to remember that the control period is set arbitrarily, different from PWM inverters. Thus, the broadband current control system is realized by Class-G power amplifiers.

This paper is organized as follows. In Section II, classification of typical power amplifiers is introduced. Topology, operation, power efficiency, and noise characteristics of Class-G power amplifier are described in Section III. In Section IV, the broadband current control of a linear motor with Class-G power amplifiers is explained. The simulations of the current controllers using Class-D and Class-G power amplifiers are mentioned in Section V. Furthermore, the experimental setup and results of the Class-G power amplifiers and a linear motor is represented in Section VI. The last section concludes this paper.

II. CLASS-B AND CLASS-D POWER AMPLIFIERS

In this section, classification of typical Class-B and Class-D power amplifiers is introduced.

A. Class-B Power Amplifier

Fig. 1(a) represents circuit diagram of general Class-B power amplifier. The input voltage v_{in} is amplified by the first stage voltage amplifier. After that, second stage source follower which consists of MOSFETs, drive a load. In the Class-B operation, the MOSFETs are equivalent to a sort of a resister, which consumes power calculated as

$$P_D = V_{DS}I_D \quad (1)$$

where P_D , V_{DS} , and I_D denote power loss of the FET, voltage between drain and source, and drain current. In general, power efficiency of Class-B push-pull amplifier η_b is 78.5%

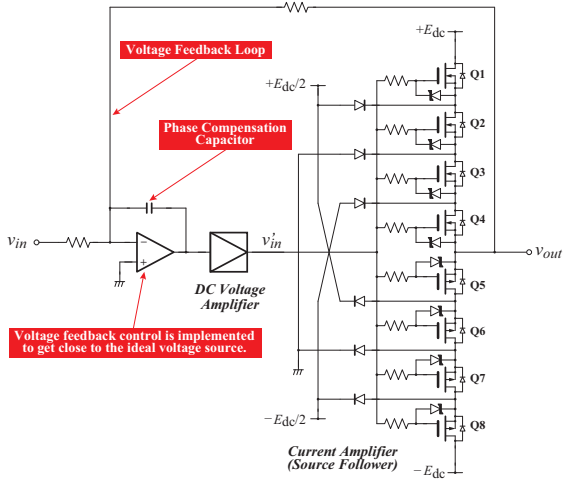


Fig. 2. 4-series Class-G power amplifier with voltage feedback.

on condition that phase factor = 1 [14]. By using Class-B amplifier, current noise can be reduced. However, Class-B amplifier is not suitable for driving high-power actuators from view point of energy-saving.

B. Class-D Power Amplifier

Fig. 1(b) shows circuit diagram of general Class-D power amplifier. In the Class-D power amplifier, PWM is used in order to drive the load. According to eq. (1), power losses of FETs are approximately zero, because either V_{DS} or I_D is equal to zero in steady state. In general, power efficiency of Class-D amplifier is approximately over 90 %. Therefore, Class-D amplifier which is equivalent to PWM or PDM inverter is very useful for applications of high-power motor drive. However, the output current includes the noise caused by the modulation. Furthermore, sampling and update period is fixed to PWM carrier. Hence, Class-D amplifier is not suitable for motion control.

III. CLASS-G POWER AMPLIFIER

A. Topology and Operation

Fig. 2 shows circuit diagram of Class-G power amplifier. Class-G power amplifier consists of FETs, gate resistors, diodes, and Zener diodes. E_{dc} represents the dc link voltage. Fundamentally, Class-G amplifier is operated as source follower. Therefore, voltage amplification is almost 1. Hence, it is necessary to implement the voltage amplifier in front of the leg.

The operation of Class-G power amplifier is as follows:

- $0 < v'_{in} < \frac{+E_{dc}}{2}$
Q1, Q2, Q3, and Q4 are set as off-state, active state, on-state, and on-state, respectively.
- $v'_{in} = \frac{+E_{dc}}{2}$
Q1, Q2, Q3, and Q4 are set as off-state, on-state, on-state, and on-state, respectively.
- $\frac{+E_{dc}}{2} < v'_{in} < +E_{dc}$
Q1, Q2, Q3, and Q4 are active-state, on-state, on-state, and on-state, respectively.

TABLE I
POWER EFFICIENCY VS. NUMBER OF SERIES.

Number of series n	4	6	8	10	12	14
Efficiency η_g [%]	84.2	87.7	89.9	91.4	92.5	93.4

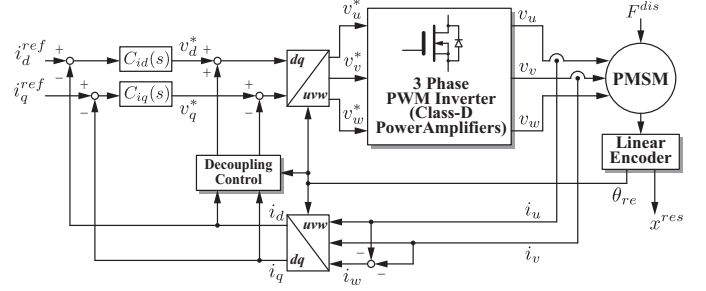


Fig. 3. Block diagram of current control system with Class-D power amplifiers.

- $v'_{in} = +E_{dc}$

All high side FETs Q1 to Q4 are turned into on-state.

By this method, Class-G power amplifier is able to drive the load according to the input voltage v'_{in} .

B. Power Efficiency

The theoretical power efficiency of Class-G amplifier η_g is derived as [14]

$$\eta_g = \frac{\pi n^2}{16 \sum_{k=1}^{\frac{n}{2}} \sqrt{nk - k^2}} \quad (2)$$

where n denotes number of series (= half of the total number of FETs). TABLE I represents relation between the power efficiency and the number of series. According to eq. (2), the number of series should be set as large from view point of the power efficiency.

C. Noise Characteristics

In Class-D amplifier, voltage signal is modulated by PWM or PDM in order to generate desired voltage as mentioned in Section II. On the other hand, PWM or PDM are unused in the case of Class-G amplifier. Therefore, the noise caused by PWM carrier is not generated.

IV. BROADBAND CURRENT CONTROL

In this section, broadband current control system is explained. At first, conventional current control using Class-D power amplifier is mentioned. Secondly, this paper describes FPGA-based control using Class-G power amplifiers, which is enabled to output fine current quickly.

A. Current Control System

Firstly, the current sensors detect the current of the linear motor i_u, i_v . At the same time, the encoder counter measures the position response and the electrical angle of the linear motor x^{res}, θ_{re} . The current responses in 3-phase are transformed into the current responses in d-q coordinate i_d, i_q . In order to

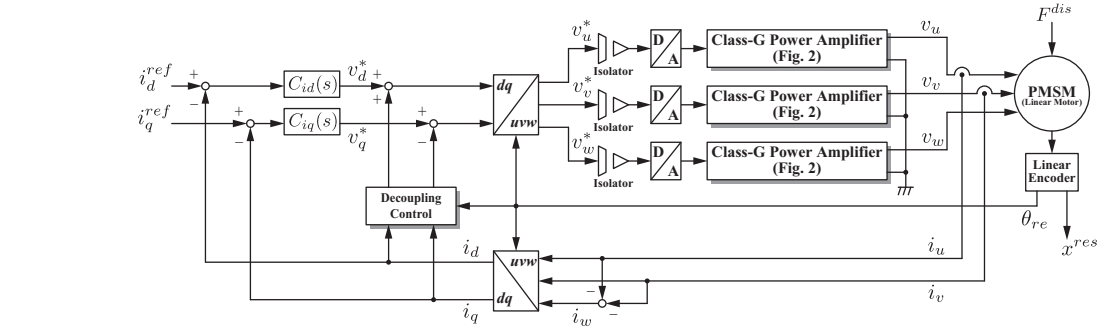


Fig. 5. Block diagram of current control system with Class-G power amplifiers.

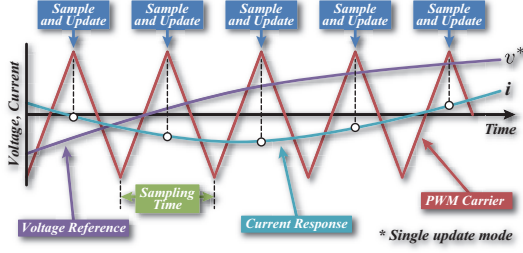


Fig. 4. Operation of current control with PWM inverter, which is classified as Class-D power amplifier.

achieve current control, the voltage references v_d^* , v_q^* in d-q coordinate are calculated as follows:

$$\begin{bmatrix} v_d^* \\ v_q^* \end{bmatrix} = C_i \begin{bmatrix} i_d^{ref} - i_d \\ i_q^{ref} - i_q \end{bmatrix} \quad (3)$$

where i_d^{ref} , i_q^{ref} denote the current command in d-q coordinate. C_i means the current controllers expressed as

$$C_i = \text{diag}(C_{id}(s), C_{iq}(s)) \quad (4)$$

where $C_{id}(s)$ and $C_{iq}(s)$ represents the current controller on the d-axis and the q-axis, respectively. In this paper, classical PI controllers are utilized in order to achieve the current control. Additionally, the linear motor used in this research is classified as surface permanent magnet synchronous motor (SPMSM). Therefore, the controllers are described as follows:

$$C_{id}(s) = C_{iq}(s) = K_{ip} + \frac{K_{ii}}{s} \quad (5)$$

where K_{ip} and K_{ii} represent the proportional gain and the integral gain, respectively. The each gain of the PI regulators are determined as follows:

$$K_{ip} = \omega_c L_a \quad (6)$$

$$K_{ii} = \omega_c R_a \quad (7)$$

where L_a and R_a mean inductance and the resistance of the linear motor. ω_c denotes bandwidth of the current control.

Furthermore, the voltage references should be modified, because interference of d-axis and q-axis occurs. The modified

voltage references v_d^{*l} , v_q^{*l} are calculated as

$$\begin{bmatrix} v_d^{*l} \\ v_q^{*l} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -L_a \omega_{re} \\ 0 & 1 & -L_a \omega_{re} & 0 \end{bmatrix} \begin{bmatrix} v_d^* \\ v_q^* \\ i_d \\ i_q \end{bmatrix} + \begin{bmatrix} 0 \\ e_q \end{bmatrix} \quad (8)$$

where ω_{re} and e_q represent electrical angular velocity and reverse voltage in q-axis. The voltage references in d-q coordinate are transformed into the 3-phase voltage references v_u^* , v_v^* , v_w^* .

B. Motor Drive using Class-D Amplifiers

Fig. 3 represents the current control system using Class-D power amplifiers. The power amplifiers drive the linear motor according to the 3-phase voltage references calculated in the previous section. To generate PWM gate pulse, the voltage references are compared to PWM carrier shown in Fig. 4. Finally, switching operation of MOSFET or IGBT is controlled by the gate pulse.

As shown in Fig. 4, update time of the voltage signal is limited because of PWM switching operation. Furthermore, switching noise occurs due to the PWM carrier in the Class-D power amplifier. Therefore, in general, sampling of the current responses is executed at the edge of the triangular PWM carrier, as shown in Fig. 4. In short, the update time and the sampling time are restricted by the period of the PWM carrier. The frequency of the PWM carrier is limited to switching speed and loss of MOSFET or IGBT. Hence, high-speed current control using Class-D power amplifier cannot be realized.

C. Motor Drive using Class-G Amplifiers

Class-G power amplifiers are utilized in this paper. Fig. 5 shows the current control system using Class-G power amplifiers. Fundamentally, the control method is same as (3)–(8). The power amplifiers drive the linear motor depending on the 3-phase voltage references, which are calculated as (8). However, as shown in Fig. 5, Class-G power amplifier is used instead of general PWM inverter. The advantages of Class-G power amplifier are as follows:

- High-power efficiency,
- Low noise current,
- Short update time and sampling time.

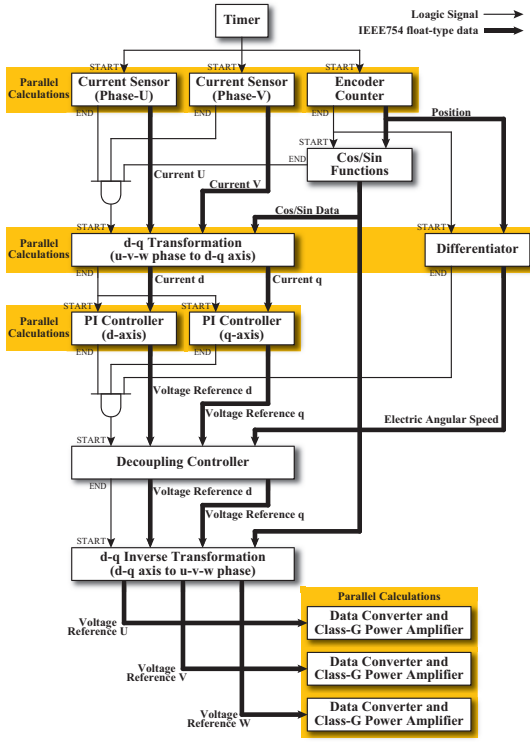


Fig. 6. Schematic diagram of FPGA design for broadband current control system.

The second and third advantages are enabled to realize a high-speed current control. In particular, the third one is a very important element for broadband current control.

As mentioned above, in the general PWM inverter, the update time of the voltage signal and the sampling time of the current responses are fixed to PWM carrier. Even if high-speed arithmetic is carried out by using FPGA, the total control period is limited. On the other hand, the update time and the sampling time is able to be set arbitrarily, because Class-G power amplifier is sort of a linear amplifier. Hence, broadband current control is able to be realized by using Class-G power amplifiers.

For advanced motion control, the bandwidth of the current control ω_c is a very important parameter because it is inner loop. The bandwidth ω_c should be set as large as possible, in order to realize quick and fine motion control. By using the wideband current control with Class-G power amplifiers, performance of motion control will be enhanced.

D. FPGA-based Current Control

In this paper, FPGA is utilized for high-speed processing of the current control. Fig. 6 depicts the schematic diagram of FPGA design for the broadband current control system. When the start signal(START) is detected, the each entities calculate (3)–(8). The calculation is carried out in parallel, so that computation time is short. In the process, the data type is single precision floating point which standard is IEEE754. After that, the entities output the complete signal(END) when

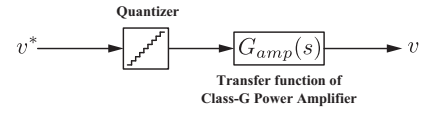
TABLE II
PARAMETERS IN NUMERICAL SIMULATIONS AND EXPERIMENT.

Parameters	Symbols	Value	Units
Resistance of linear motor*	R_{uv}	9.0	Ω
Inductance of linear motor*	L_{uv}	1.3	mH
Force constant	K_{fn}	4.3	N/A
Mass of actuator	M	0.15	kg
Encoder resolution	x_{reso}	100	nm
Resolution of D/A converters (Class-G)	v_{reso}	8.61	mV
Sampling time (Class-D)	T_s	50	μ s
Sampling time (Class-G)	T_s	8	μ s
Carrier frequency of PWM (Class-D)	$f_c = \frac{1}{T_s}$	20	kHz
Bandwidth of current controllers (case 1)	ω_c	6280	rad/s
Bandwidth of current controllers (case 2)	ω_c	25100	rad/s

* Average value of u-v, v-w, and w-u lines.



(a) Actual block diagram.



(b) Block diagram for simulations.

Fig. 7. Block diagram of Class-G Amplifier in numerical simulations.

the individual process is done. Finally, the voltage references are transmitted to Class-G power amplifiers.

V. SIMULATIONS

The numerical simulations of the current control system with the Class-D and Class-G power amplifiers are conducted.

A. Simulation Conditions

TABLE II shows the parameters of the plant systems and the controllers. The control period T_s in the simulation of the Class-D power amplifiers is set to 50 μ s, due to the loss and switching time of the devices such as MOSFET or IGBT. On the other hand, the control period of the Class-G power amplifiers is set to 8 μ s. In the simulations, the bandwidth of the current control ω_c is set to 6280 rad/s (case 1), and 25100 rad/s (case 2).

For the precise numerical calculations, this paper assumes that the Class-G power amplifier shown in Fig. 7 is expressed as

$$G_{amp}(s) = \frac{g_{amp}}{s + g_{amp}} e^{-T_s s}. \quad (9)$$

g_{amp} denotes bandwidth of the Class-G power amplifier, and is determined by a pre-experiment.

B. Simulation Results

Figs. 8 and 9 represent simulation results of the current control with Class-D power amplifiers. In case 1, the q-axis current is normally controlled. However, the actuator system runs out of control in case 2.

Figs. 10 and 11 show simulation results of the current control with Class-G power amplifiers. By using the Class-G power amplifiers, the current controller is normally operated even where $\omega_c = 25100$ rad/s (case 2). As shown in Fig. 11, the overshoot occurs. The overshoot is caused by sort of time delay, which is represented as (9).

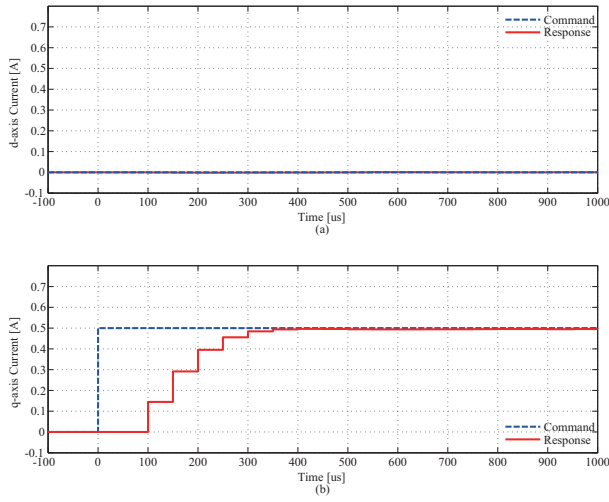


Fig. 8. Simulation results of current control (case 1). (Class-D amplifiers, $\omega_c = 6280$ rad/s $\simeq 1$ kHz) (a) d-axis Current i_d^{ref} , i_d . (b) q-axis Current i_q^{ref} , i_q .

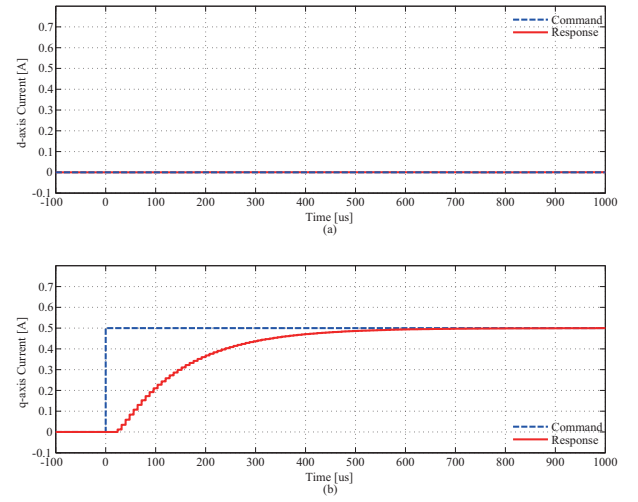


Fig. 10. Simulation results of current control (case 1). (Class-G amplifiers, $\omega_c = 6280$ rad/s $\simeq 1$ kHz) (a) d-axis Current i_d^{ref} , i_d . (b) q-axis Current i_q^{ref} , i_q .

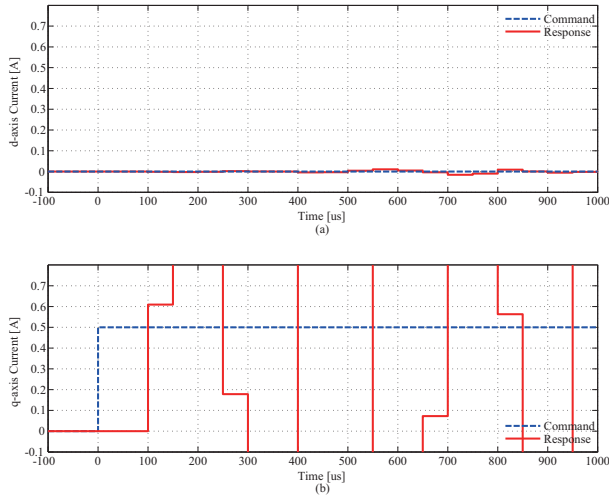


Fig. 9. Simulation results of current control (case 2). (Class-D amplifiers, $\omega_c = 25100$ rad/s $\simeq 4$ kHz) (a) d-axis Current i_d^{ref} , i_d . (b) q-axis Current i_q^{ref} , i_q .

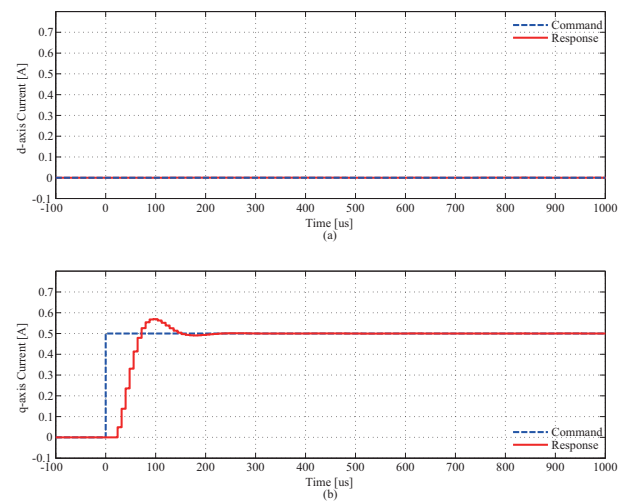


Fig. 11. Simulation results of current control (case 2). (Class-G amplifiers, $\omega_c = 25100$ rad/s $\simeq 4$ kHz) (a) d-axis Current i_d^{ref} , i_d . (b) q-axis Current i_q^{ref} , i_q .

VI. EXPERIMENT

In this research, the experiment is carried out to evaluate performances of the current control system with the Class-G power amplifiers.

A. Experimental Setup

Fig. 12 represents the experimental equipment, which consists of the 4-series Class-G power amplifiers and FPGA. In the equipment, n-channel and p-channel MOSFETs are 2SK1530 and 2SJ201, which are complementary type. The FPGA is ALTERA EP3C55F780C8N which has 55856 Logic Elements(LE) and 156 multipliers. For implementing the current control system, the usage of LEs is about 60%. In this experiment, the current of a SPMSM linear motor is controlled. TABLE II shows the setup parameters. This paper

conducts two experiment; first experiment is case 1, other one is case 2. In case 1, the bandwidth of the current controllers is set as 6280 rad/s ($\simeq 1$ kHz). The bandwidth is set as 25100 rad/s ($\simeq 4$ kHz) in case 2.

B. Experimental Results

Fig. 13 represents experimental results of the current control with the Class-G power amplifiers. Of course, the current of the linear motor is able to be controlled in case 1.

Fig. 14 shows experimental results of the Class-G power amplifiers. Though the overshoot occurs as with the simulation results shown in Fig. 11, the current controllers are normally operated even where the bandwidth is set to 25100 rad/s ($\simeq 4$ kHz).

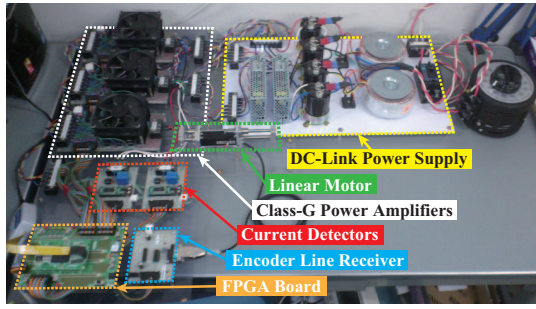


Fig. 12. Experimental setup.

VII. CONCLUSION

In this paper, the broadband current control system using FPGA and the Class-G power amplifiers was realized. The current controllers with general PWM inverter, which is classified as Class-D power amplifier, did not operate in case that the bandwidth of current control is enlarged. On the contrary, the current control system using Class-G power amplifiers was able to be operated normally, even where the bandwidth was set as 25100 rad/s ($\simeq 4\text{kHz}$). By the numerical simulations and the experiment, validity of the current controllers with Class-G power amplifiers was verified.

The bandwidth of current control, which is implemented generally as inner loop, should be enlarged for fast motion control. Thus, the broadband current control realized in this paper is useful for fast motion control. In particular, the high-performance current control will be applied to servo systems in the field of real-world haptics as mentioned in introduction.

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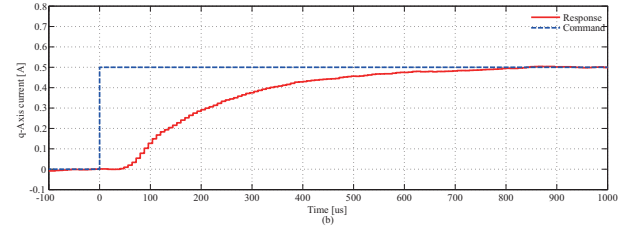
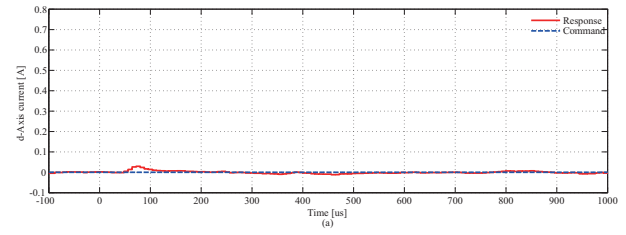


Fig. 13. Experimental results of current control (case 1). (Class-G amplifiers, $\omega_c = 6280 \text{ rad/s} \simeq 1 \text{ kHz}$) (a) d-axis Current i_d^{ref} , i_d . (b) q-axis Current i_q^{ref} , i_q .

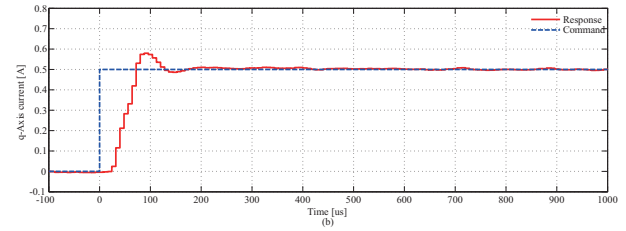
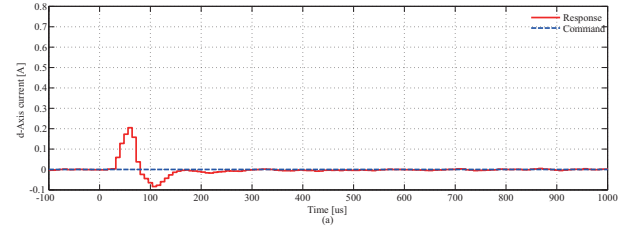


Fig. 14. Experimental results of current control (case 2). (Class-G amplifiers, $\omega_c = 25100 \text{ rad/s} \simeq 4 \text{ kHz}$) (a) d-axis Current i_d^{ref} , i_d . (b) q-axis Current i_q^{ref} , i_q .